Quiz 4 — Instructor Key

Friday, October 9, 2026

1. (4 points)

(a) 1. Exponentials and polynomials are continuous, and the denominator at 0 is $2 \neq 0$.

(b) 7. Factor $x^2 - x - 12 = (x-4)(x+3)$ and cancel for $x \neq 4$.

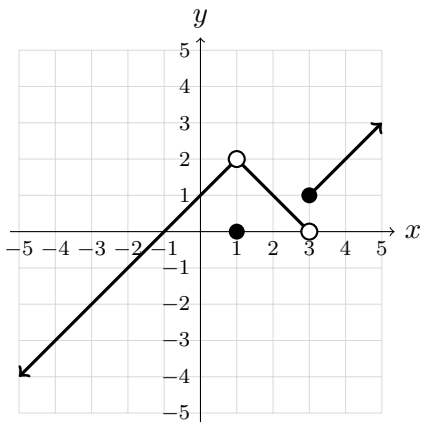
(c) 3. For large positive x,

$$\sqrt{x^2+6x} - x = \frac{6x}{\sqrt{x^2+6x}+x} = \frac{6}{\sqrt{1+6/x}+1} \longrightarrow 3.$$

(d) Left: $-\infty$; right: $+\infty$; two-sided: DNE.

2. (2 points) The left limit and $H(2)$ equal $2k-1$, and the right limit is 7. Thus $2k-1=7$ gives $k=4$. All three quantities then agree.

3. (4 points) (a) Graph:



(b)

a	$g(a)$	$\lim_{x\to a^-} g(x)$	$\lim_{x\to a^+} g(x)$	$\lim_{x\to a} g(x)$
1	0	2	2	2
3	1	0	1	DNE

(c) At 1, g is not continuous: its limit is $2 \neq g(1)$. This is removable; set $g(1)=2$. At 3, the one-sided limits disagree, so g has a jump and is not continuous. Changing only $g(3)$ cannot repair it.

Grading: award credit for correct reasoning when a minor arithmetic error carries forward. In 1(d), split credit across the one-sided limits and the two-sided conclusion. In 3(a), give half credit for the segments and half for endpoint markings; in 3(b), give half credit per row.